

Results

Classification Reports

(Color Indices)

Random Forest Classifier(with redshift)

Classification Report :

Test Accuracy: 0.9935968725777643

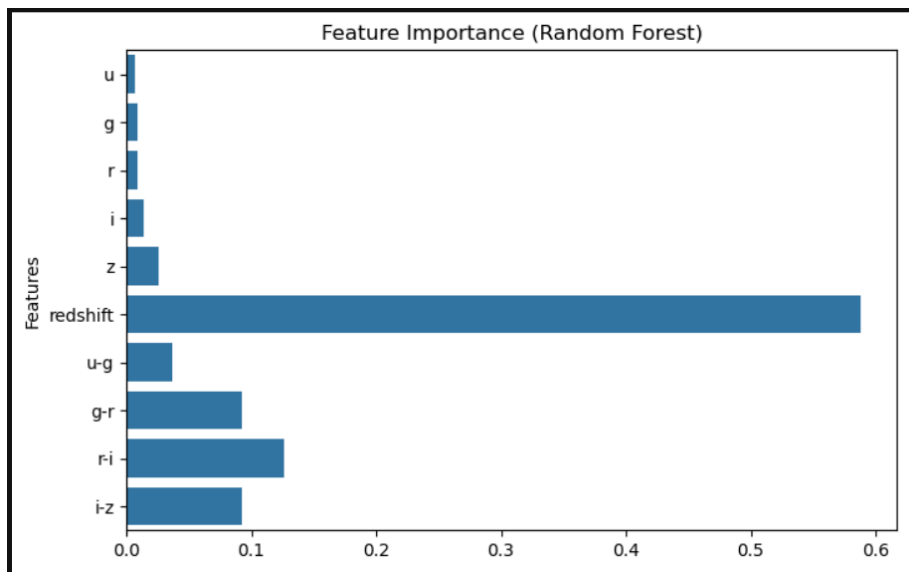
Confusion Matrix:

```
[[17523   38    17]
 [  135 2631     0]
 [     0     0 9329]]
```

Classification Report:

	precision	recall	f1-score	support
0	0.99	1.00	0.99	17578
1	0.99	0.95	0.97	2766
2	1.00	1.00	1.00	9329
accuracy			0.99	29673
macro avg	0.99	0.98	0.99	29673
weighted avg	0.99	0.99	0.99	29673

Feature Importance:

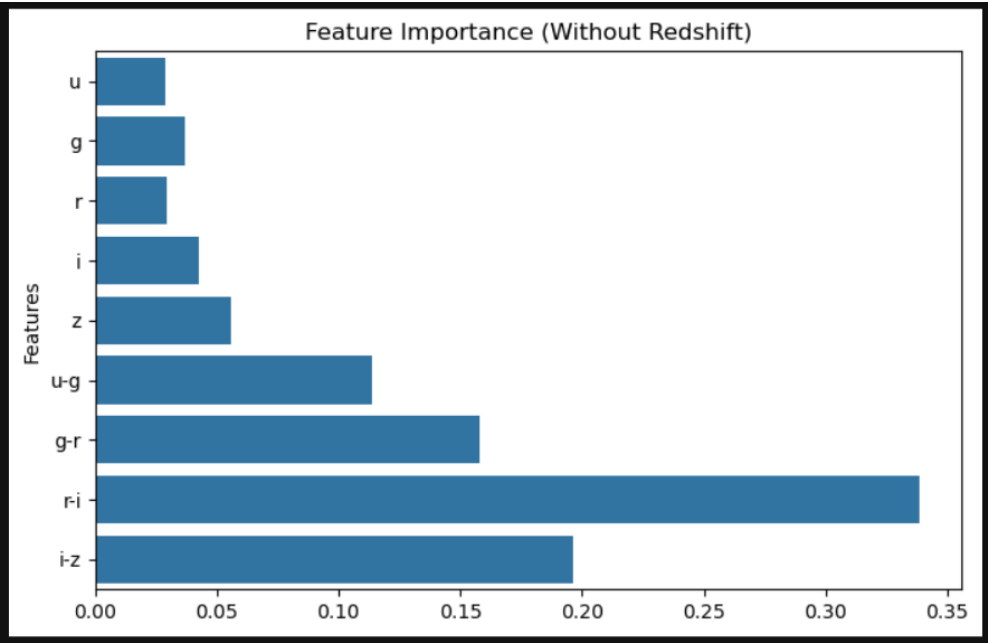


Random Forest classifier (without Redshift)

Classification Report:

Test Accuracy: 0.9291611903076871					
Confusion Matrix:					
[[17182 93 303]					
[159 2267 340]					
[903 304 8122]]					
Classification Report:					
	precision	recall	f1-score	support	
0	0.94	0.98	0.96	17578	
1	0.85	0.82	0.83	2766	
2	0.93	0.87	0.90	9329	
accuracy			0.93	29673	
macro avg	0.91	0.89	0.90	29673	
weighted avg	0.93	0.93	0.93	29673	

Feature Importance :



XGB Classifier (with Redshift)

Classification Report:

Accuracy: 0.993866477942911

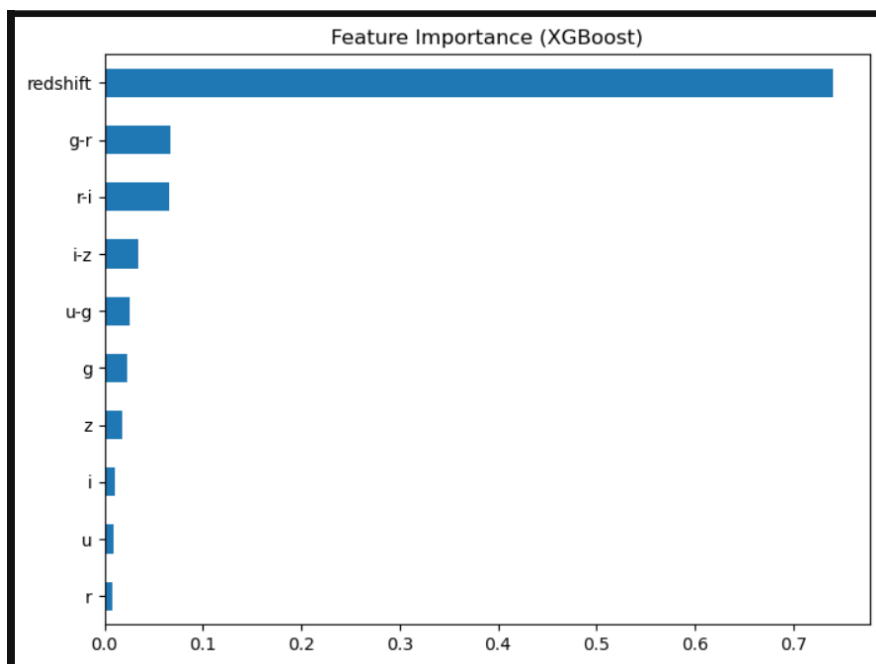
Confusion Matrix:

```
[[17521   34    23]
 [  124 2642     0]
 [     1     0 9328]]
```

Classification Report:

	precision	recall	f1-score	support
0	0.99	1.00	0.99	17578
1	0.99	0.96	0.97	2766
2	1.00	1.00	1.00	9329
accuracy			0.99	29673
macro avg	0.99	0.98	0.99	29673
weighted avg	0.99	0.99	0.99	29673

Feature Importance :



XGB classifier (without Redshift)

Classification report :

```
Accuracy: 0.9326997607252384
```

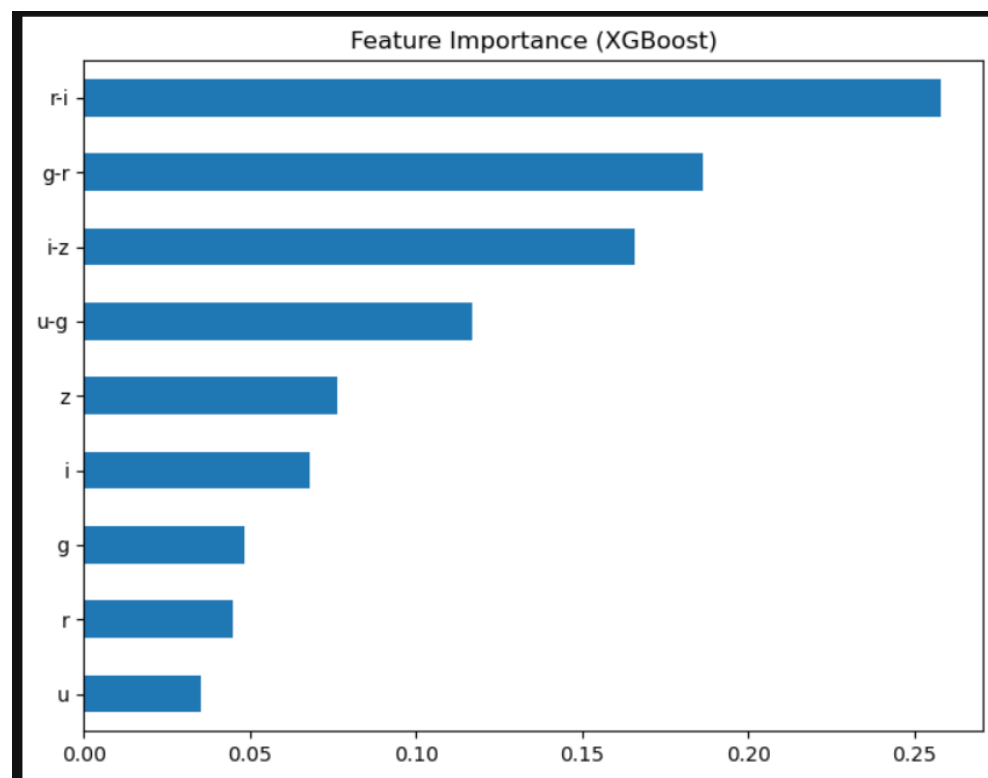
Confusion Matrix:

```
[[17134   91   353]
 [  151 2278   337]
 [   776  289 8264]]
```

Classification Report:

	precision	recall	f1-score	support
0	0.95	0.97	0.96	17578
1	0.86	0.82	0.84	2766
2	0.92	0.89	0.90	9329
accuracy			0.93	29673
macro avg	0.91	0.89	0.90	29673
weighted avg	0.93	0.93	0.93	29673

Feature Importance:



FNN Keras (with Redshift)

Classification Report :

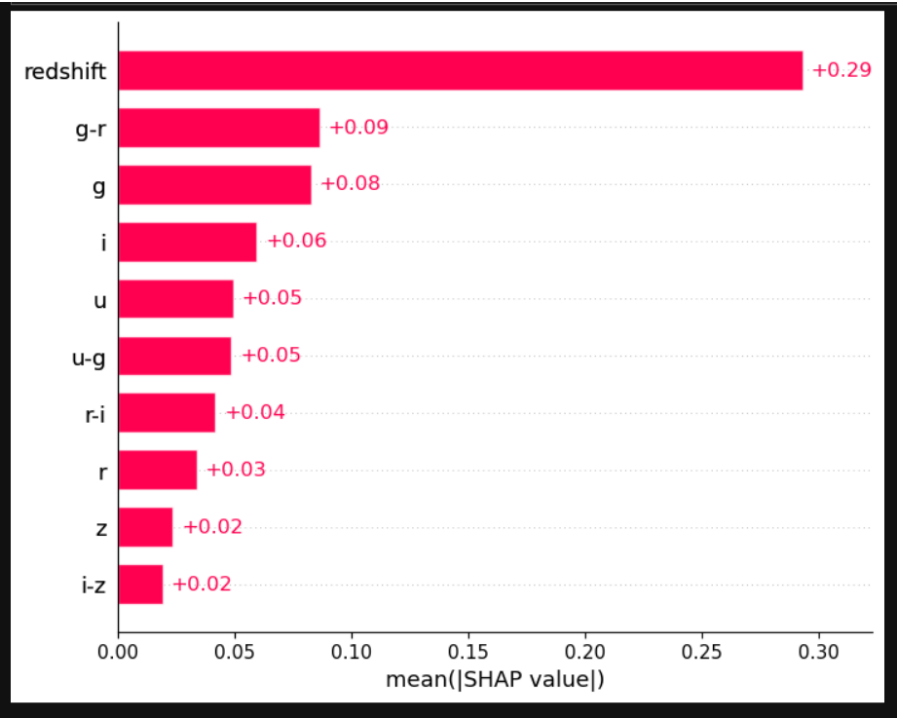
Test Accuracy: 0.9883058667182922
928/928 1s 1ms/step

Confusion Matrix:
[[17351 179 48]
[88 2678 0]
[32 0 9297]]

Classification Report:

	precision	recall	f1-score	support
0	0.99	0.99	0.99	17578
1	0.94	0.97	0.95	2766
2	0.99	1.00	1.00	9329
accuracy			0.99	29673
macro avg	0.98	0.98	0.98	29673
weighted avg	0.99	0.99	0.99	29673

Feature Importance :



FNN keras (without Redshift):

Classification Report:

Test Accuracy: 0.9221851229667664

928/928 1s 1ms/step

Confusion Matrix:

[[16762 293 523]

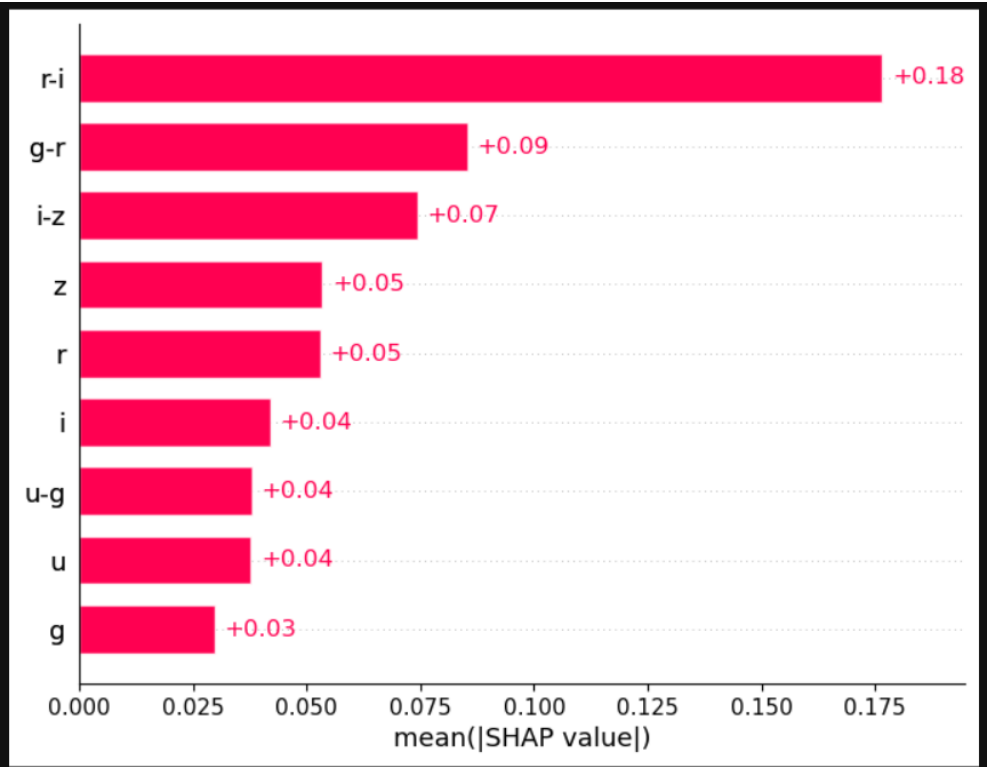
[100 2476 190]

[622 581 8126]]

Classification Report:

	precision	recall	f1-score	support
0	0.96	0.95	0.96	17578
1	0.74	0.90	0.81	2766
2	0.92	0.87	0.89	9329
accuracy			0.92	29673
macro avg	0.87	0.91	0.89	29673
weighted avg	0.93	0.92	0.92	29673

Feature Importance:



Stacking (FNN + XGB) (with Redshift)

Classification Report :

```
K-Fold Stacking Accuracy: 0.9939001786135544
```

```
Confusion Matrix:
```

```
[[17509   48   21]
 [  108 2658    0]
 [    4    0 9325]]
```

```
Classification Report:
```

	precision	recall	f1-score	support
0	0.99	1.00	0.99	17578
1	0.98	0.96	0.97	2766
2	1.00	1.00	1.00	9329
accuracy			0.99	29673
macro avg	0.99	0.99	0.99	29673
weighted avg	0.99	0.99	0.99	29673

Agreement :

```
XGB Accuracy: 0.9939001786135544
FNN Accuracy: 0.9909682202675834
Stacking Accuracy: 0.9939001786135544
```

```
Agreement %: 0.9942371853199878
Accuracy (agree): 0.9952884550199986
Accuracy (disagree): 0.7543859649122807
```

Stacking (FNN+ XGB) (without Redshift)

Classification Report:

K-Fold Stacking Accuracy: 0.934115188892259

Confusion Matrix:

```
[[17081  104  393]
 [  136 2319  311]
 [   694  317 8318]]
```

Classification Report:

	precision	recall	f1-score	support
0	0.95	0.97	0.96	17578
1	0.85	0.84	0.84	2766
2	0.92	0.89	0.91	9329
accuracy			0.93	29673
macro avg	0.91	0.90	0.90	29673
weighted avg	0.93	0.93	0.93	29673

Agreement :

XGB Accuracy: 0.9322616520068749

FNN Accuracy: 0.9255552185488491

Stacking Accuracy: 0.934115188892259

Agreement %: 0.9568294409058741

Accuracy (agree): 0.9495632572555649

Accuracy (disagree): 0.5917252146760343